

The Genus of the Fermat Curve

the topological foundation of the cyclotomic stands

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1 Setting

The Fermat curve $C_N = \{x^N + y^N = z^N\} \subset \mathbb{P}^2$ is the basic object of the cyclotomic stands. Its topology is determined by a single integer N , and the first pipeline step — the genus g — is purely topological: no analysis, no spectral theory. The genus determines the dimension of the space of holomorphic forms, the size of the period matrix, and ultimately the scale of the whole construction.

Lemma 1 (genus of the Fermat curve). *Let $N \geq 4$. The projection $\pi_x: C_N \rightarrow \mathbb{P}^1$ is a morphism of degree N , unramified over ∞ and ramified exactly over the N points ζ_N^k , with ramification index N at each $P_k = (\zeta_N^k, 0)$. The genus equals $g = \frac{(N-1)(N-2)}{2}$; in particular $g(15) = 91$, $g(30) = 406$.*

Proof. Over any $x_0 \in \mathbb{P}^1$ the equation $y^N = 1 - x_0^N$ has exactly N solutions with multiplicity, hence $\deg \pi_x = N$. At P_k the local expansion $1 - x^N = -N\zeta_N^{k(N-1)}(x - \zeta_N^k) + \dots$ gives $y^N = c(x - \zeta_N^k)$ with $c \neq 0$: the parameter is y and $x - \zeta_N^k \sim c^{-1}y^N$, so the ramification index is N . Over ∞ there is no ramification: at $z = 0$ the equation $X^N + Y^N = 0$ has N distinct solutions with unramified projection. The Riemann–Hurwitz formula gives $2g - 2 = -2N + N(N-1) = N^2 - 3N$, hence $g = (N-1)(N-2)/2$: 91 and 406 at $N = 15, 30$. $\square$

Remark 2. The same Plücker formula for a smooth plane curve of degree m gives $\frac{(m-1)(m-2)}{2}$ — the agreement of the two readings serves as a built-in control: at $N = 4$ both give 3 (the Klein quartic is also a genus-3 curve, but a *different* curve).

Protocol data: numbers

$g(15) = 91$; period matrix 91×182 ; $g(30) = 406$; period matrix 406×812 ; for comparison: $g(4) = 3$, $g(7) = 15$, $g(22) = 231$.

Summary of results: summary

The genus is proved from Riemann–Hurwitz with all components explicit. Consequences: the character censuses $91 = 1 + 6 + 84$ and $406 = 1 + 6 + 9 + 30 + 84 + 276$ agree with the genera (check V1, exact integer arithmetic).

Verification: reproduction

`python3 laboratory.py` → “Stand N=15/30 → V1 census”; Lean: `genus_fermat` in `HodgeLaboratory.lean` — machine proofs of $(15-1)(15-2)/2 = 91$ and $(30-1)(30-2)/2 = 406$.